Pràctica 2.

BLE, provisioning de la Wi-Fi, restabliment i telemetria REST

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Sistemes Encastats

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Resultats

Exercici 1 · Servidor GATT propi

UUID Servei de Configuració: 9bd46fc4-d6ab-4d64-81ee-1c0d3be13a3d
UUID Característica credencials: 1009d074-8913-4371-8313-99958c49c265
UUID Característica estat: 7e93a750-5e54-41a7-9532-8315e7921b63

credencials: Propietat d'escriptura (PROPERTY_WRITE), rep una cadena JSON amb el ssid i la pass.
estat: Propietats de lectura i notificació (PROPERTY_READ | PROPERTY_NOTIFY), publica l'estat actual en text pla.

Exercici 2 · Provisioning de la Wi-Fi

Per conectar cal enviar la següent comande per BLE::
{"ssid":"NomWi-Fi","pass":"Contrasenya"}

```
Attempting Wi-Fi connection (1/3)...
Connexió completada en 5531 ms
```

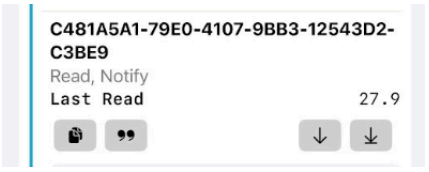
Després de reiniciar

```
E (16) i2c.master: I2C transaction unexpected nack detected
Attempting Wi-Fi connection (1/3)...
```

Exercici 3 · Restabliment de fàbrica

Si pressionem la pantalla del M5 durant 5 s, fem un restabliment de fàbrica, per poder tornar a configurar el nostre Wifi.

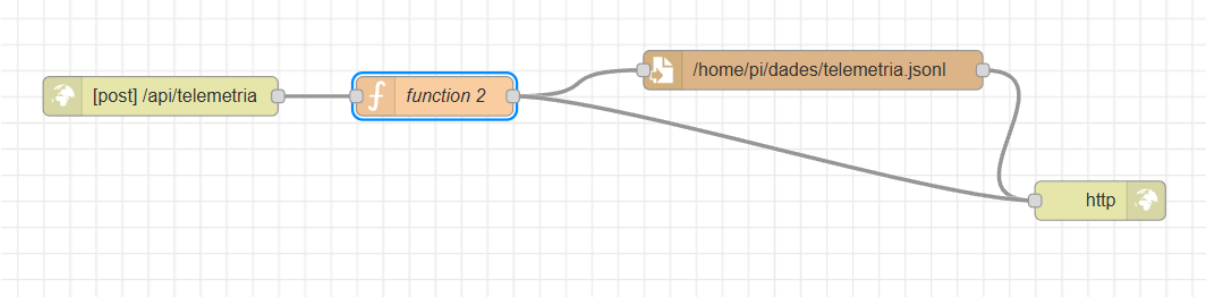
Exercici 4 · Enviament de la temperatura al mòbil per BLE



Es pot observar com rebem per BLE la temperatura, en aquest cas 27,9

Exercici 5 · Publicació de la temperatura a Node-RED per API REST

Creem el següent flux al Node-RED:

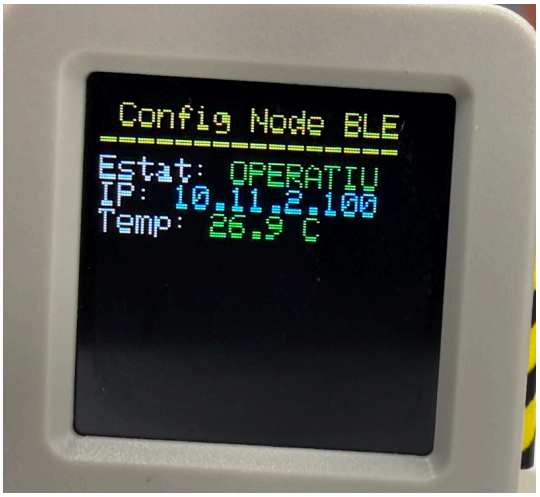


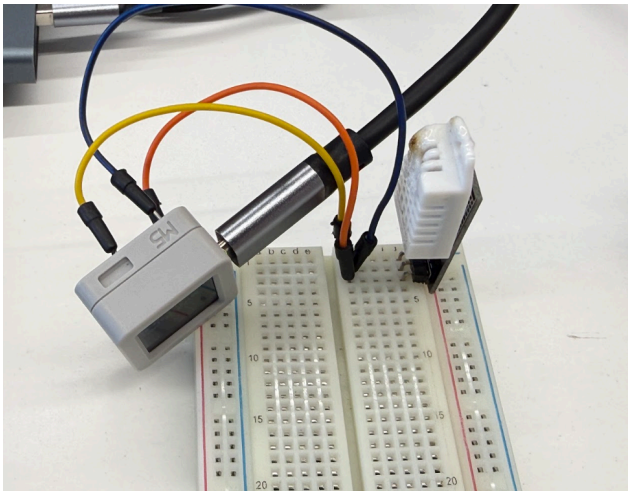
Desde ssh podem veure que efectivament funciona, i es guarden les dades.
Per veure les dades guardades utilitzem: tail -f /home/pi/dades/telemetria.jsonl

```
admin@Eric:~ $ tail -f /home/pi/dades/telemetria.jsonl
{"node_id":"ESP32_Eric","temperatura":26.9,"humitat":65.4,"rssi":-73,"uptime":10,"timestamp":"2026-10-07T14:51:21.413Z"}
{"node_id":"ESP32_Eric","temperatura":26.9,"humitat":64.4,"rssi":-65,"uptime":20,"timestamp":"2026-10-07T14:51:31.439Z"}
{"node_id":"ESP32_Eric","temperatura":26.9,"humitat":64.4,"rssi":-55,"uptime":30,"timestamp":"2026-10-07T14:51:41.479Z"}
{"node_id":"ESP32_Eric","temperatura":26.9,"humitat":64.1,"rssi":-56,"uptime":40,"timestamp":"2026-10-07T14:51:51.415Z"}
{"node_id":"ESP32_Eric","temperatura":26.9,"humitat":64,"rssi":-56,"uptime":50,"timestamp":"2026-10-07T14:52:01.547Z"}
{"node_id":"ESP32_Eric","temperatura":26.9,"humitat":63.8,"rssi":-56,"uptime":60,"timestamp":"2026-10-07T14:52:11.486Z"}
{"node_id":"ESP32_Eric","temperatura":26.9,"humitat":63.9,"rssi":-57,"uptime":70,"timestamp":"2026-10-07T14:52:21.515Z"}
{"node_id":"ESP32_Eric","temperatura":26.9,"humitat":64.3,"rssi":-53,"uptime":80,"timestamp":"2026-10-07T14:52:31.649Z"}
```

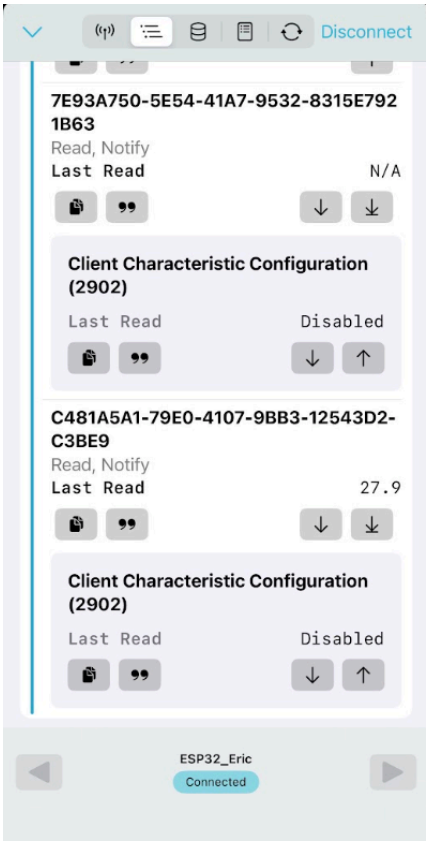
De la mateixa manera que pel serial d'Arduino podem observar si hi ha algun error o s'envia tot correctament.

```
HTTP Response code: 201
HTTP Response code: 201
HTTP Response code: 201
HTTP Response code: 201
HTTP Response code: 201
HTTP Response code: 201
HTTP Response code: 201
```





Per BLE continuem mantenint la comunicació, i podem observar la temperatura



Repositori Git: https://github.com/eric-vila/Repositori-Sistemes-Encastats/tree/main